

Directional Simulation

```
library(spatstat.geom)
```

```
## Loading required package: spatstat.data
```

```
## Loading required package: spatstat.univar
```

```
## spatstat.univar 3.1-6
```

```
## spatstat.geom 3.7-0
```

```
library(spatstat.random)
```

```
## spatstat.random 3.4-4
```

```
W <- owin(xrange = c(0, 10), yrange = c(0, 10))
```

```
sigma_parallel <- 0.60
```

```
sigma_perp <- 0.12
```

```
theta0 <- pi / 4
```

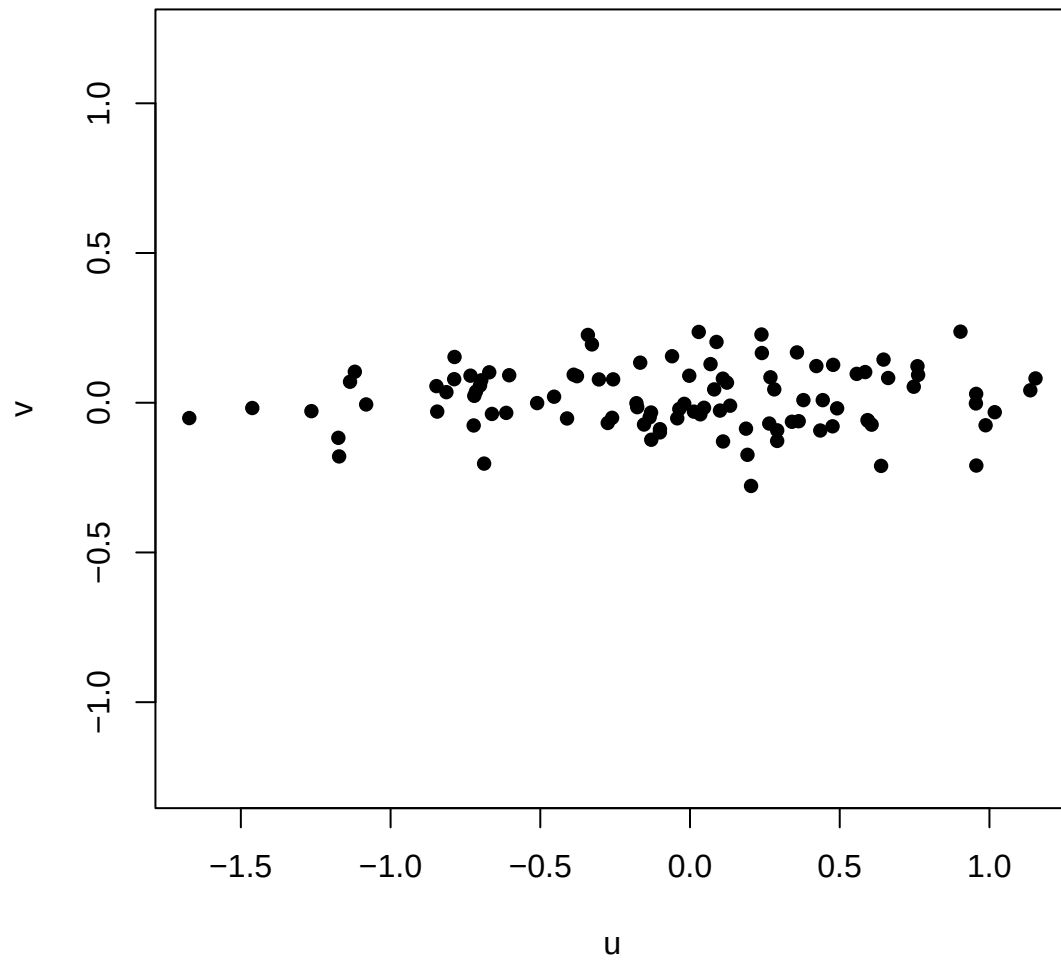
```
n <- 100
```

```
u <- rnorm(n, mean = 0, sd = sigma_parallel)
```

```
v <- rnorm(n, mean = 0, sd = sigma_perp)
```

```
plot(u, v, asp = 1, pch = 16,  
      xlab = "u", ylab = "v", main = "Before rotation")
```

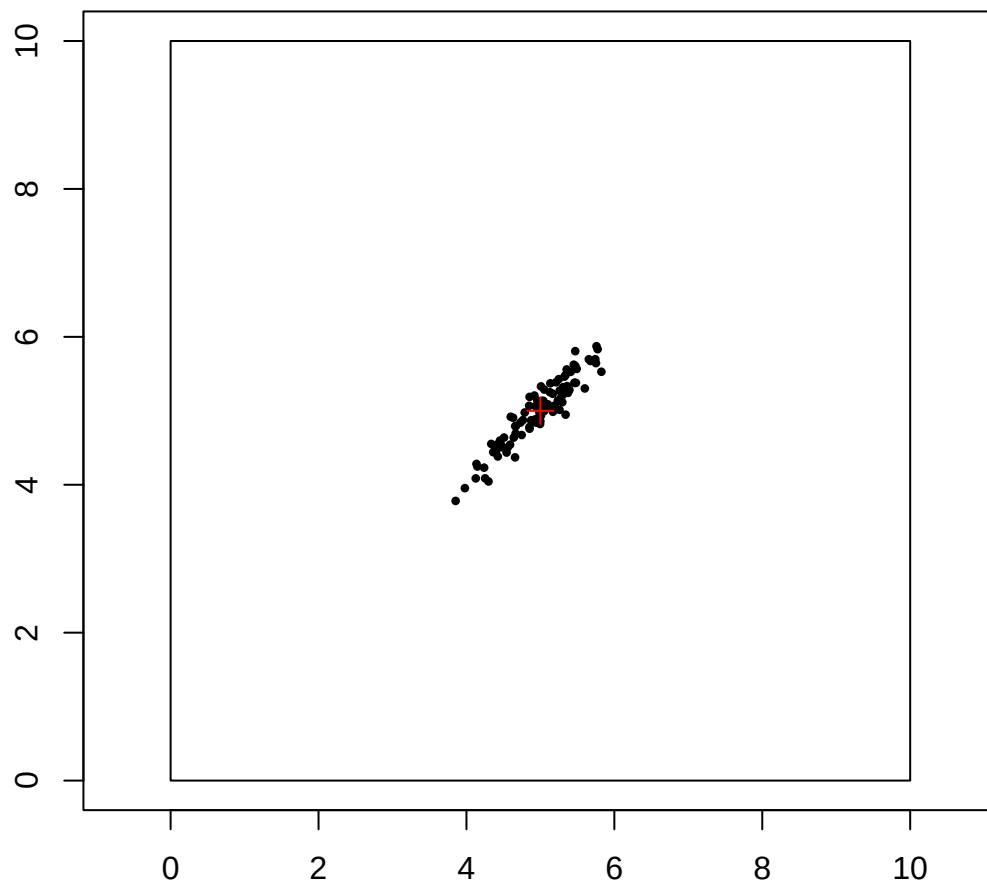
Before rotation



```
dx <- u * cos(theta0) - v * sin(theta0)
dy <- u * sin(theta0) + v * cos(theta0)
xy <- data.frame(x = 5 + dx, y = 5 + dy)

plot(W, main = "One ellipse cluster", axes = TRUE)
points(xy$x, xy$y, pch = 16, cex = 0.6)
points(5, 5, pch = 3, col = "red", cex = 1.3)
```

One ellipse cluster



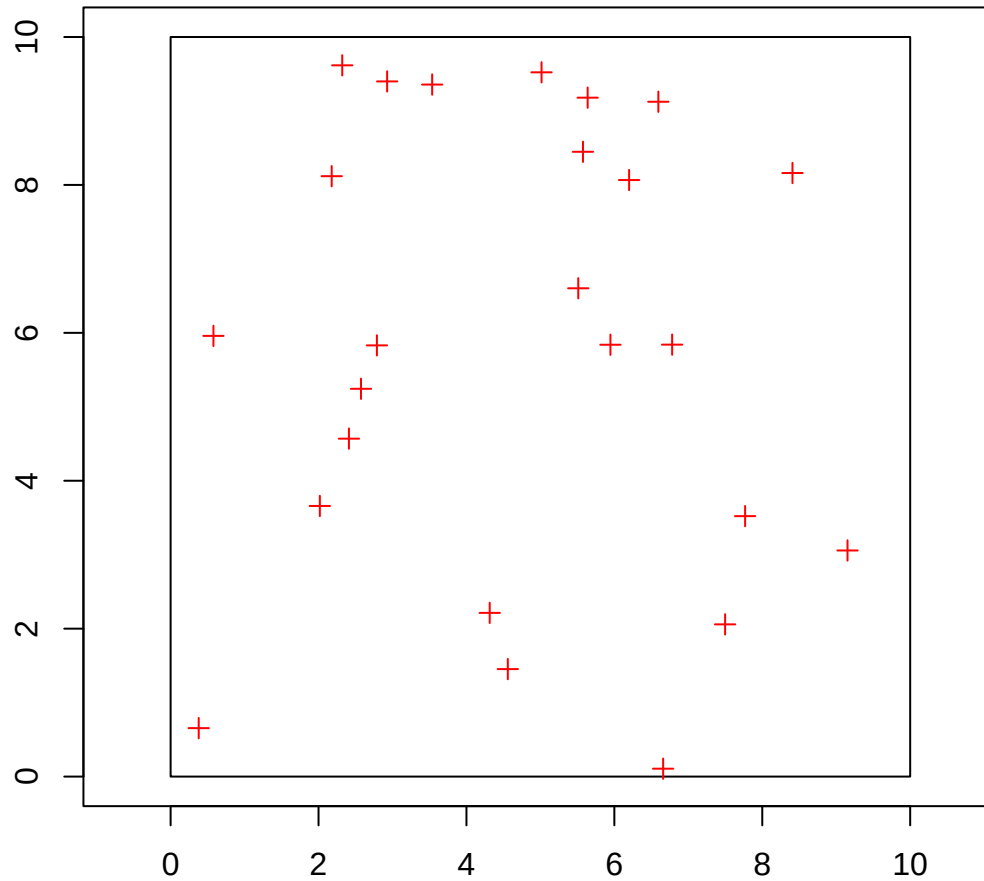
```
lambda_centers <- 0.2
X_parent <- rpoispp(lambda_centers, win = W)
xy_parent <- data.frame(x = X_parent$x, y = X_parent$y)
n_parent <- npoints(X_parent)

n_parent
```

```
## [1] 25
```

```
plot(X_parent, main = "Cluster centres", axes = TRUE,
     pch = 3, cols = "red")
```

Cluster centres



```
lambda_offspring <- 40
n_offspring <- rpois(n_parent, lambda = lambda_offspring)

n_offspring
```

```
## [1] 33 46 31 34 45 46 49 50 49 31 42 48 37 41 43 45 47 25 31 35 42 33 40 31 41
```

```
xy <- data.frame(x = numeric(), y = numeric()) #
for (i in seq_len(n_parent)) {
  u <- rnorm(n_offspring[i], sd = sigma_parallel)
  v <- rnorm(n_offspring[i], sd = sigma_perp)

  xy_i <- data.frame(
    x = xy_parent$x[i] + u * cos(theta0) - v * sin(theta0),
    y = xy_parent$y[i] + u * sin(theta0) + v * cos(theta0)
  )
  xy <- rbind(xy, xy_i)
}

keep <- inside.owin(xy$x, xy$y, W) #only keep points in the window
X <- ppp(x = xy$x[keep], y = xy$y[keep], window = W)
```

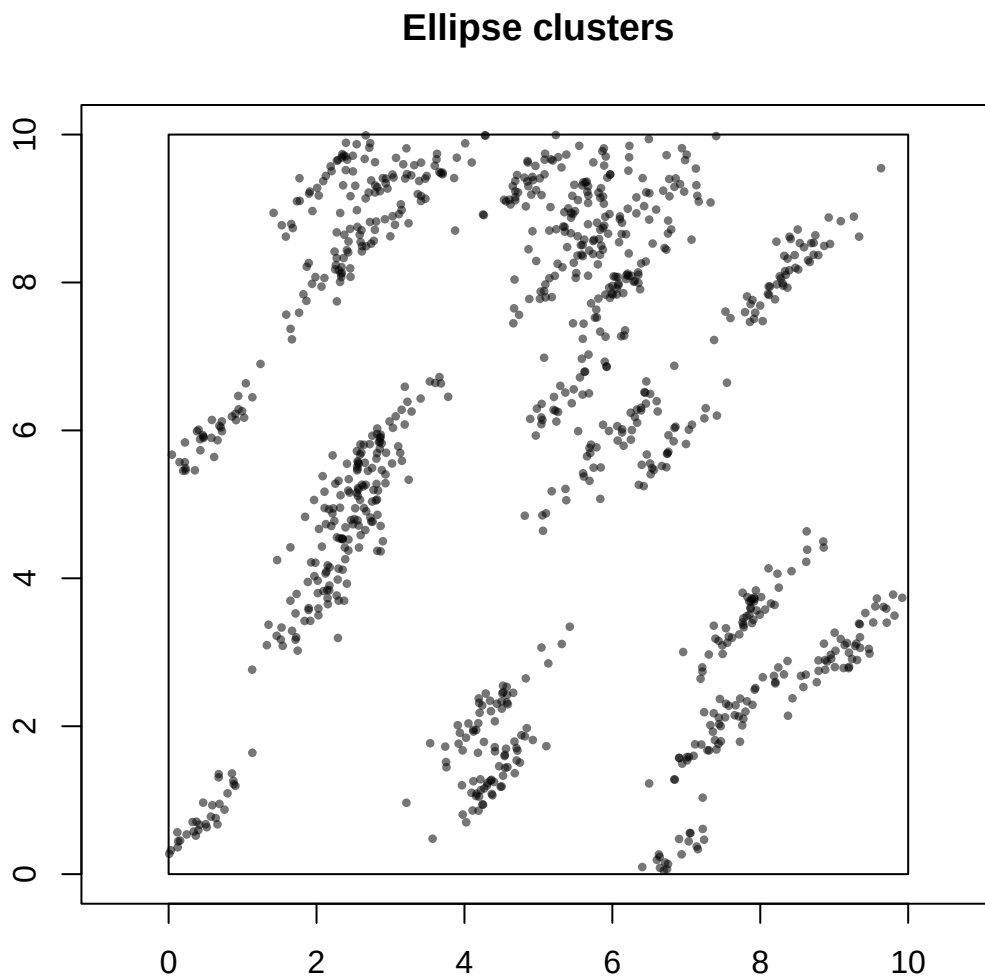
```
sum(n_offspring)
```

```
## [1] 995
```

```
npoints(X)
```

```
## [1] 951
```

```
plot(X, main = "Ellipse clusters", axes = TRUE, pch = 16, cex = 0.6)
```



```
plot(X, main = "Points and centres", axes = TRUE, pch = 16, cex = 0.6)  
plot(X_parent, add = TRUE, pch = 3, cols = "red", cex = 1.2)
```

Points and centres

